

Norbert Frydrysiak

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EDUCATION

Warsaw University of Technology

Warsaw, Poland

Master of Science in Engineering, Data Science

02.2026 – Present

- Relevant Coursework: Advanced Machine Learning, Deep Learning, Cloud Computing, Transformers with PyTorch

Warsaw University of Technology

Warsaw, Poland

Bachelor of Science in Engineering, Data Science

10.2022 – 02.2026

- GPA: 4.81/5.0
- Thesis: *Adapting the gips library for classification problem utilizing discriminant analysis – gipsDA*
- Relevant Coursework: Automated Machine Learning, Stochastic Processes, Methods of Computational Statistics, Computational Intelligence Methods, Financial Mathematics

EXPERIENCE

Teaching Assistant

02.2026 – Present

Warsaw University of Technology

Warsaw, Poland

- Conducting lab classes in "Data Warehousing & Business Intelligence Systems" and "Structured Data Processing"
- Teaching two 15-person groups of 1st and 3rd-year Data Science students for a total of 6 hours per week

Risk Model Validation Intern

07.2025 – 08.2025

Velo Bank S.A.

Warsaw, Poland

- Co-developing an AutoML package in Python for monitoring machine learning models
- Implementing modules for model stress testing, visualizations, and performance analysis in production environments
- Collaborating within a 3-person team, utilizing Git for version control and repository management

Data Science Intern in the Property and Personal Insurance Tariff Team

07.2024 – 09.2024

Generali Poland

Warsaw, Poland

- Preprocessing and analyzing travel insurance claims data, including outlier detection, missing value imputation, and dimensionality reduction
- Developing supervised learning models for claim frequency and average damage utilizing Generalized Linear Models with Poisson and Gamma distributions

PROJECTS

Computational Intelligence | *Python, NumPy, pandas, Matplotlib, Seaborn*

- Implementing Genetic Algorithms, Multi-Layer Perceptrons, and Kohonen's Self-Organizing Maps entirely from scratch using NumPy
- Developing a custom neural network framework featuring backpropagation, Momentum and RMSProp optimization, and L1 regularization for classification tasks
- Applying evolutionary algorithms and unsupervised clustering techniques to solve geometric optimization problems and visualize high-dimensional data

Hyperparameter Tunability AutoML | *Python, scikit-learn, scikit-optimize, SciPy, pandas*

- Analyzing the hyperparameter tunability of Gradient Boosting, Random Forest, KNN, and Logistic Regression models, adapting methodology from the paper "Tunability: Importance of Hyperparameters of Machine Learning Algorithms"
- Evaluating and comparing uniform sampling techniques with Bayesian optimization to determine the stability and efficiency of the tuning process
- Conducting comprehensive statistical analysis and visualizing optimization trajectories to assess the impact of specific hyperparameters on model performance

TECHNICAL SKILLS

Languages: Python, R, SQL, Java, C/C++, LaTeX, Bash

Data Science & ML: NumPy, Pandas, Scikit-Learn, Scikit-Optimize, XGBoost, PyTorch, SciPy, Matplotlib, Seaborn

Tools & Technologies: Git, Docker, Jupyter, Linux (Debian, Arch), Neovim, REST, Web Scraping

Knowledge & Concepts: Machine Learning, Reinforcement Learning, Stochastic Processes, Statistics, Linear Algebra